A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 18

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 2 points below. **1** Doubling the distance from 1 m to 2 m reduces the dose rate from 400 to 100 (one quarter) **1** Distance x 10 (1 to 10 m) reduces it by 100 times (400 to 4) – dose is proportional to 1/distance^2 **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
01.2	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 400 / 20^2 = 1 μSv h^{-1} **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (μSv h^{-1}) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1

Do not write outside the box

Question	Marking guidance	Marks
01.3	**1 mark each** 1 mark for each of the 4 points below. **1** Sealed container prevents contact / contamination so exposure stops when the worker moves away – container may absorb alpha completely – remote handling **1** Barriers (lead, concrete, thick steel flask walls) absorb radiation before it reaches the worker – thickness depends on radiation type **1** Dose received is proportional to time near the source so work is organised to minimise time close to it (job rotation) **1** Dosemeters / film badges record exposure – air monitors detect alpha-emitting particles – contamination monitors on leaving check that no material is carried out **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** Stable iodine saturates the thyroid **1** So radioactive iodine-131 released is not taken up / concentrated in the thyroid gland **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	2
01.5	1 mark for the point below. **1** Exposure should be As Low As Reasonably Achievable **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 302-303; AQA spec 3.4.3.2.14.	
02.1	**1 mark each** 1 mark for each of the 3 points below. **1** Solar PV cost fell dramatically from about 480 to 50 $/MWh (about 90%), the steepest fall **1** Wind fell steadily from 200 to 45 **1** Gas stayed roughly constant at 60-70 – solar and wind became cheaper than gas around 2015-2020 and are projected to fall further **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
02.2	**1 mark each** 1 mark for each of the 3 points below. **1** Development / research costs are still being paid for and there are no economies of scale in manufacture **1** They compete with established technologies whose development costs were paid long ago and which benefit from mass production and existing infrastructure **1** As output grows, mass production, improved designs / efficiency and experience reduce costs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Most of the renewable cost is the initial equipment, paid for with loans on which interest must be paid – running costs are very low **1** Most of the gas project's cost is running costs (fuel) paid from income without borrowing, so it has lower interest payments **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	2
02.4	**1 mark each** 1 mark for each of the 2 points below. **1** Environmental damage (acid rain damage to buildings, crops, forests; health effects of air pollution; climate change) is paid for by others (health service, agriculture) not the energy user **1** Some impacts (climate change) cannot yet be costed – decommissioning and waste costs are also uncertain **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **decommissioning** → safely dismantling a power station at the end of its life **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 179-181; AQA spec 3.3.2-3.3.3.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Deforestation:** less interception so less evaporation from leaves – less transpiration so less water returned to the atmosphere and less precipitation downwind – more infiltration and surface runoff, higher river flows / flooding **2 Urban development:** impermeable concrete / tarmac reduce infiltration and aquifer recharge – runoff is faster and greater, increasing river flooding downstream – drains carry water quickly to rivers **2 Agriculture:** soil compaction by machinery / trampling reduces infiltration and soil moisture, increases runoff – irrigation increases evaporation – loss of soil biota (worms) reduces infiltration and water retention – drainage lowers water tables **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **transpiration** → water lost as vapour from the leaves • **infiltration** → water soaking into the soil surface • **impermeable** → does not let water through **Do not credit** • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss	6
03.2	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Higher temperatures increase evaporation and change the amount, timing, type (rain vs snow) and location of precipitation **1** More rapid melting of snow and ice changes river regimes: more flooding after rain, lower flows in dry periods **1** Changed wind patterns move humid air to different areas **1** Sea level rise / salt incursion into coastal aquifers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Most water abstracted is cleaned and returned to the river / sea so the main impact is on quality (pollution) not quantity **1** Effects are noticeable only where large urban centres draw on restricted local supplies (aquifers, small rivers) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 116-117; AQA spec 3.2.2.1.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**Marks shared** Up to 6 marks in total - share them between the 4 points below according to how completely each one is made. **up to 2** Detritivores (worms, beetle larvae, millipedes, woodlice) physically break up dead organic matter, increasing its surface area, and release nutrients – worms create tunnels improving drainage and aeration **up to 2** Decomposers (bacteria, fungi) secrete enzymes that break down dead organic matter, releasing nutrient ions (ammonification) and forming humus which binds particles into crumb peds **up to 2** Nitrogen-fixing bacteria convert nitrogen gas to ammonium ions – nitrifying bacteria oxidise ammonium to nitrite then nitrate for root absorption **up to 2** Mycorrhizal fungi form symbiotic associations with roots increasing phosphate uptake **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ammonification** → ammonium released from dead organic matter and protein • **humus** → dark, decomposed organic matter in soil • **peds** → soil crumbs / aggregates of soil particles	6
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** This is the range of tolerance of most plants and soil biota – acidic soils leach nutrients and mobilise toxic aluminium, damaging root membranes **1** In alkaline soils phosphates become insoluble / unavailable **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **range of tolerance** → the range of conditions in which a species can survive	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Interstitial spaces are reduced so aeration falls (aerobic processes such as nitrification and decomposition slow) and roots cannot penetrate **1** Infiltration falls so waterlogging is more likely and runoff / erosion increase **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Interstitial** → pore spaces / air spaces between the particles • **nitrification** → ammonium oxidised to nitrite and then nitrate • **Infiltration** → water soaking into the soil surface	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-161; AQA spec 3.2.5.1.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Image databases:** identifying individuals (tiger stripes, whale shark spots, dolphin fins) to find territory size, movements, lifespan, social groups - few species have unique recognisable features **2** eDNA: detecting the presence of a species from DNA shed into water (great crested newt, invasive carp) - shows presence not abundance – contamination / DNA transport by water **2 Geolocator tags:** tracking small birds on migration where transmitters are too heavy - only record light and time so position is approximate – animal must be recaptured to download data **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	6
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Microwaves emitted by the satellite reflect off the surface giving data on wave height, wind velocity, sea surface altitude and oil slicks (which smooth the surface) **1** Microwaves pass through cloud so monitoring is not interrupted by weather **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
05.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1 Droppings:** diet, gender, territory size **1 Owl pellets:** diet **1** Feeding marks (chewed hazel nuts - dormouse), tracks / footprints, nests / burrows, scratching posts (territory) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-407; AQA spec 3.7.2.3.	
06.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** Aerobic decomposition produces little methane and returns nutrients to the soil	1

Do not write outside the box

Question	Marking guidance	Marks
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** It may contain concentrated heavy metals / toxic residues **1** It is stored in solid form at high pH / in hazardous-waste landfill so the metals stay insoluble **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2
06.3	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Impermeable polymer / clay liner to stop leachate escaping **1** Collection and treatment of leachate **1** Collection of methane and use as fuel **1** Daily covering of waste / final capping and restoration **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Impermeable** → does not let water through • **leachate** → polluted liquid draining out of waste **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 279-285; AQA spec 3.4.3.2.12.	
07.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (210 - 12) / 210 x 100 **1** = 94.3% (accept 94%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. • **recycling** → reprocessing a material to make something new **Do not credit** • A bare wrong answer with no working. • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new	2

Do not write outside the box

Question	Marking guidance	Marks
07.2	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Conserves mineral reserves / extends their lifetime **1** Reduces the environmental impacts of mining (land take, habitat loss, spoil, dust, noise, turbid water) **1** Reduces mineral processing pollution (eg red mud from bauxite refining) **1** Reduces waste disposal impacts / landfill volume **1** Less CO2 / climate change **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **recycling** → reprocessing a material to make something new **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	3
07.3	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1 Identification of materials:** slow, labour-intensive to determine composition where it matters (eg alloys, plastic types) **1 Separation of mixed materials:** alloys cannot easily be separated so recycled aluminium cannot be used where pure metal is needed – mixed wastes reduce quality **1** Reduction in quality / contamination (eg paper mixed with food waste) **1 Transport / collection:** small dispersed quantities cannot use bulk transport so costs and energy per tonne are higher (may exceed savings) **1** Lack of consumer cooperation in separating wastes – waste losses to litter and general waste – labour costs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **recycling** → reprocessing a material to make something new **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new	4
07.4	1 mark for the point below. **1** Designing products so that the materials can be re-used at the end of their lives - easy separation of components and identification of materials - extending the lifetime of mineral use **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 146-147; AQA spec 3.2.3.7.	

Do not write outside the box

Question	Marking guidance	Marks
08.1	**1 mark each** 1 mark for each of the 4 points below. **1** Trade winds blow westwards across the Pacific near the equator, driving a surface current in the same direction **1** Surface water moving away from South America is replaced by deep, cold water drawn upwards (upwelling) near the coast **1** The upwelling water is rich in nutrients (nitrates, phosphates) so large algal / phytoplankton blooms occur **1** These support a rich food web including the anchovy fishery – the water warms as it travels west **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	4
08.2	**1 mark each** 1 mark for each of the 3 points below. **1** The trade winds weaken or reverse so the surface current slows / reverses and warm water spreads east **1** Upwelling of nutrient-rich cold water off South America stops **1** Phytoplankton productivity collapses so the food web / anchovy population collapses – catch fell from about 7.7 to 1.7 million tonnes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
08.3	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Heavy rain and floods in the normally dry coastal deserts of Peru / South America **1** Drought and reduced rainfall in eastern Australia / Indonesia **1** Droughts in NE Africa, southern Africa, China **1** Fewer hurricanes in the North Atlantic / fewer cyclones in Japan **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
08.4	1 mark for the point below. **1** Events appear to be occurring more frequently and are driven by wind patterns that warming may alter, but the causes of ENSO are not fully understood and natural fluctuations make trends hard to separate **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 96-99; AQA spec 3.2.1.2, 3.2.2.3.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**1 mark each** 1 mark for each of the 3 points below. **1 Mutagen:** an agent that changes the chemical structure / base sequence of DNA (causes mutations), eg ionising radiation, UV, PCBs, cadmium, asbestos **1 Carcinogen:** a mutagen that causes cancer / uncontrolled cell division producing a tumour **1 Teratogen:** an agent that causes birth abnormalities by preventing normal gene expression (inhibiting proteins / enzymes) without changing DNA structure, eg mercury, 2,4,5-T **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **ionising radiation** → radiation with enough energy to remove electrons and form ions **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	3
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** Teratogens do not change the DNA structure, so the abnormality is not in the genes and cannot be passed on **1** Mutagens change DNA – if the mutation is in a gamete / egg / sperm (gonadic effect) it can be inherited **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
09.3	**1 mark each** 1 mark for each of the 2 points below. **1 Somatic:** mutation in a general body cell, affecting the individual only (eg cancer / tumour) **1 Gonadic:** mutation in cells of the ovaries or testes / gametes / embryo, which may cause abnormalities in offspring **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
09.4	**1 mark each** 1 mark for each of the 3 points below. **1** Two or more pollutants whose effects interact to produce a different (usually more serious) effect **1** The pollutants themselves do not react to form a new pollutant (that would be a secondary pollutant) **1 Example:** ozone damages leaf cuticles so sulfur dioxide can cause more damage to the exposed cells – smoke and sulfur dioxide in smogs – neonicotinoids with fungicides **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any appropriate named example, including ones that are not listed here. • **synergism** → two pollutants together doing more damage than the sum of each alone	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 239-240; AQA spec 3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Smoke:** solid particulates (PM10, PM5, PM1) from incomplete combustion – small particles stay suspended, penetrate lungs – toxic chemicals (acids, phenols, metals) carried on particles – high albedo cools climate – blocks light reducing photosynthesis – damages buildings • Smoke smogs form under temperature inversions in valleys (London 1952, 12 000 deaths) – synergism with SO2 • **Controls:** Clean Air Act 1956 (smokeless zones), smokeless fuels / gas / electricity, coal treatment, electrostatic precipitators, cyclones, bag filters, scrubbers, DPFs, better combustion – smoke smogs have largely been eliminated in UK cities • **Acid rain gases:** SO2 (coal combustion, smelting sulfide ores) oxidised by ozone to SO3 - sulfuric acid – NOx (high temperature combustion) - nitric acid – HCl from PVC incineration – reactive gases travel long distances (transboundary pollution) • **Effects:** pH denatures proteins / enzymes in exposed cells (gills, eggs, root hairs, stomata) – dissolves exoskeletons – leaches Ca and Mg then mobilises toxic Al and Pb – corrodes metals, dissolves limestone – severity depends on bedrock (limestone neutralises) • **Controls:** fuel desulfurisation (coal washing, hydrodesulfurisation, low-sulfur ship fuels), dry and wet FGD, fluidised bed combustion, catalytic converters, urea sprays – switching from coal to gas / renewables	9

Do not write outside the box

Question	Marking guidance	Marks
	• **Evaluation:** SO2 emissions in Europe have fallen dramatically and lake pH is recovering – NOx harder to control because of growing vehicle numbers – catalytic converters need lead-free fuel and warm engines – FGD is costly and produces wastes / gypsum – controlling primary pollutants also prevents secondary pollutants (ozone, PANs) • Judgement on relative success of controls for particulates vs SO2 vs NOx **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **particulates** → fine solid particles / smoke and dust • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back • **synergism** → two pollutants together doing more damage than the sum of each alone **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	
	Model answer - Level 3 (8 marks)	
	Smoke is particulate matter from incomplete combustion of coal, diesel, wood and wastes. Its effects follow from particle size: PM10 and smaller particles stay suspended, travel far and penetrate the lungs, carrying acids, phenols and metals; smoke also reflects sunlight (cooling), blocks light for photosynthesis and blackens buildings. Under a temperature inversion in a valley it accumulates as smog, and it acts synergistically with sulfur dioxide - the London smog of 1952 killed about 12 000 people. Acid rain gases are different: sulfur dioxide from coal combustion and sulfide smelting, and nitrogen oxides from high-temperature combustion, are reactive and soluble, forming sulfuric and nitric acids that travel long distances across borders. Low pH denatures proteins in exposed cells such as gills, eggs and root hairs, dissolves exoskeletons and limestone, corrodes metals and leaches calcium and magnesium from soils while mobilising toxic aluminium, with severity depending on bedrock.	
	Smoke control has been the clearest success: the Clean Air Act's smokeless zones, the switch to gas and electricity, electrostatic precipitators, cyclones, bag filters and diesel particulate filters have virtually eliminated smoke smogs in developed countries. Sulfur dioxide has also fallen dramatically through fuel desulfurisation, flue-gas desulfurisation and the move from coal to gas and renewables, and lake pH is recovering, although FGD is costly and produces waste. Nitrogen oxides remain the hardest: catalytic converters and fluidised-bed combustion work, but converters need unleaded fuel and warm engines and vehicle numbers keep rising. Controls have therefore succeeded for particulates and sulfur, only partly for nitrogen.	
	Weaker answer - Level 2 (5 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Smoke is made of small particles from burning coal and diesel. It causes breathing problems and smogs like in London in 1952. Acid rain is caused by sulfur dioxide and nitrogen oxides which dissolve in rain and make it acidic. Acid rain kills fish and trees and damages buildings. Smoke is controlled by smokeless zones and filters. Acid rain is controlled by taking the sulfur out of fuel and catalytic converters in cars. These methods have reduced pollution a lot, especially smoke.	
	What makes the difference	
	• Level 3 links each effect to a property (particle size, solubility, reactivity, pH denaturing proteins); Level 2 describes effects without the mechanism.	
	• Level 3 evaluates each control separately with evidence (recovering lake pH, cost of FGD, why NOx is harder); Level 2 gives one general sentence.	
	• Both answers use the 1952 smog, but Level 3 explains the conditions (inversion, synergism with SO_2) that caused it.	
10.2	**Any 1 of 2** 1 mark each for any 1 of the 2 points below, in any order, maximum 1. **1** Reduces growth rates of trees / crops **1** Damages leaf cuticles (allowing SO2 to cause more damage) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-254; AQA spec 3.4.3.2.1-3.4.3.2.5.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss the difficulties in monitoring and predicting global climate change and evaluate the methods used to overcome them.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Nature of the problem:** interconnected atmosphere-hydrosphere-biosphere systems – changes over different time and spatial scales – natural fluctuations (solar output, orbit) masking or exaggerating anthropogenic change – time delay between cause and effect (ocean heat capacity) • **Limitations of historic data:** sparse global coverage, unsophisticated equipment, heat island bias, no atmospheric composition records • **Proxy data and their reliability:** dendrochronology, coral growth rings, pollen in dated sediments, ice cores (gas bubbles, isotope ratios, volcanic ash layers, compaction problems) • **Modern monitoring:** long-term data sets (Mauna Loa), electronic equipment, satellites (LEO vs geostationary), Argo floats, remote sensing of ice thickness / area / sea level • Feedback mechanisms and tipping points adding uncertainty – ecological uncertainty (species survival, distribution, fragmentation) • **Computer models:** how they are tested and refined – importance of accurate representative data – scenario-based predictions and ranges	25
	• Uncertainty over ENSO / Conveyor / jet stream responses • **Evaluation and conclusion:** confidence in the overall trend and human cause despite detailed uncertainty – role of peer review and international bodies (IPCC) *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 89-90, 100-104, 10; AQA spec 3.2.1.2, 3.1.1.3.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Scale of the problem:** how energy is used (transport, buildings, industry, water) – links between affluence and per capita use • **Transport:** bulk / integrated transport, ATM, vehicle design (aerodynamics, mass, tyres, KERS, combustion efficiency), low embodied energy, design for end of life, driver behaviour, electric vehicles • **Buildings:** orientation and passive solar, SA:V, thermal mass, low embodied energy materials, earth sheltering, glazing (low-e, inert gas, vacuum), insulation optimum, heat exchangers, automatic ventilation, occupancy sensors, thermostats, low-energy appliances, behaviour • **Industry:** heat recovery and exchangers, insulation and high-volume storage, CHP, integrated manufacture, recycling and mass reduction, high-voltage grid, peak shaving, ICT management • **Limits:** rebound effect, cost and payback, existing building stock, embodied energy of replacements, behaviour • Evaluation with quantitative examples – comparison with the cost of new supply	25
	– conclusion that conservation is the cheapest first step but cannot alone meet demand growth *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 170-174, 225-235; AQA spec 3.3.4.2, 3.3.1.	

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